

ASSIGNMENT #4

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Your name: _____

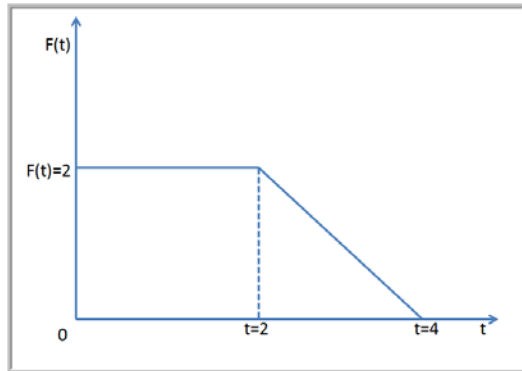
Due Date: Wednesday March 04 by 5:00pm in my office

Number of points.....100

Your score.....

Note: All problems are worth 100 points. Your final score will be the average value

Problem 4A: Determine the Laplace transform of the following input signal:



Problem 4B: Using partial fraction expansions find the inverse of the following Laplace transforms

$$Y_1(s) = \frac{s+4}{(s+1)^2(s+2)} \quad \text{and} \quad Y_2(s) = \frac{8(s+2)}{s(s^2+4)}$$

Problem 4C: Use Laplace transforms to find the solution of the following differential equation:

$$\begin{aligned} \frac{du}{dt} + 4u &= f(t) \\ t=0 \quad u &= 0 \\ f(t) &= \begin{cases} 0 & t < 0 \\ 5 & 0 \leq t < 0.2 \\ 0 & 0.2 \leq t \leq 2.0 \end{cases} \end{aligned}$$

Plot the solution from $\{t = 0 \text{ to } t = 2\}$

Problem 4D: For each of the following functions $u(s)$, what can you say about $u(t)$ $0 \leq t \leq \infty$ without solving for $u(t)$? In particular, determine whether $u(t)$ is converging or diverging; whether is smooth or oscillatory; what is the initial value $u(0)$ and the final asymptotic value $u(\infty)$

$$u(s) = \frac{s+1}{s(s+2)(s+3)}$$

$$u(s) = \frac{s+4}{s(s+2)(s^2-2s+5)}$$